AutoGenEx - Documentation

Project Drivers

1. The Purpose of the Project

1.1 Background
Typically, lecturers have an existing collection of exam questions and tasks which are recycled to determine new exams. This leads to the problem of having recurring manual work to set up new exams each semester.

1.2 Goal
AutoGenEx is about to develop an application that generates new exams from existing tasks automatically. It shall be delivered as a docker container that can be deployed at the university.

2. The Stakeholders

<u>Customer:</u> (Prof. Dr. Marc Hesenius / Lecturers of the university)
Decisions:
1. Exams as latex code
2. Web-based application (docker container)

<u>Development team:</u>

Decision	**Rationale**
Database (MongoDB)	A database is used to store, organize, and manage exam, course and task data. It ensures data is structured, secure, and easily accessible, while supporting data integrity. MongoDB is chosen for its flexibility and scalability. It stores data in JSON-like documents, allowing for dynamic and evolving data structures without the need for a fixed schema.
CLSI-Service (Overleaf)	CLSI-Service from overleaf is used because it already provides compilation of latex code into a pdf. It also provides errors if compilation fails.
GitHub (Repository)	Share central code repository among the development team and make the source code available for future work. Possibility to work on different services with several branches.

Project Constraints

3. Mandated Constraints

- Exams must be created as latex code
- The application must be web-based
- The application must be delivered as a docker container

4. Naming Conventions and Terminology

- Topic: Overall category to form a cluster of tasks

User Stories

1. As a lecturer, i want to add new tasks to the datastore, so that they can be used when generating future exams.
2. As a lecturer, i want to choose between an exam version with solutions and a version without solutions, so that i can generate both a student version and a correction version.
3. As a lecturer, i want to select a course, so that the generated exam matches the content of that course.
4. As a lecturer, i want to specify the total number of points for an exam, so that the selected tasks match this total.
5. As a lecturer, i want to adjust the number of points assigned to individual tasks, so that the total number of points is reached.
6. As a lecturer, i want to be notified when the selected tasks do not match the desired total number of points, so that i know i must adjust task points.
7. As a lecturer, i want to choose between alternative tasks, so that no task type is included more than once and the exam contains both short and long tasks.
8. As a lecturer, i want to specify topics, so that tasks are selected according to these topics.
9. As a lecturer, i want to preview the current exam design, so that i can verify its completeness and correctness before finalizing it.
10. As a lecturer, i want to add new courses and define their topics, so that exams can be generated for new courses.
11. As a lecturer, i want to download the final exam as a PDF file, so that i can store and distribute it.
12. As a lecturer, i want tasks to be selected randomly, so that different exam variants can be generated.
13. As a lecturer, i want to delete tasks, so that only relevant tasks are taken into account for exam generation.

Architecture